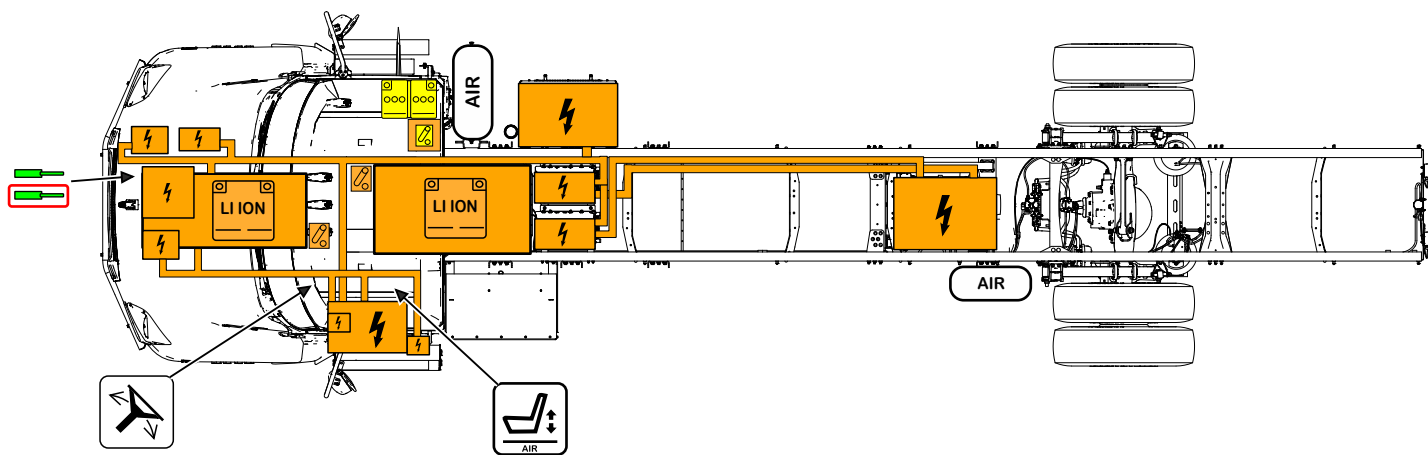
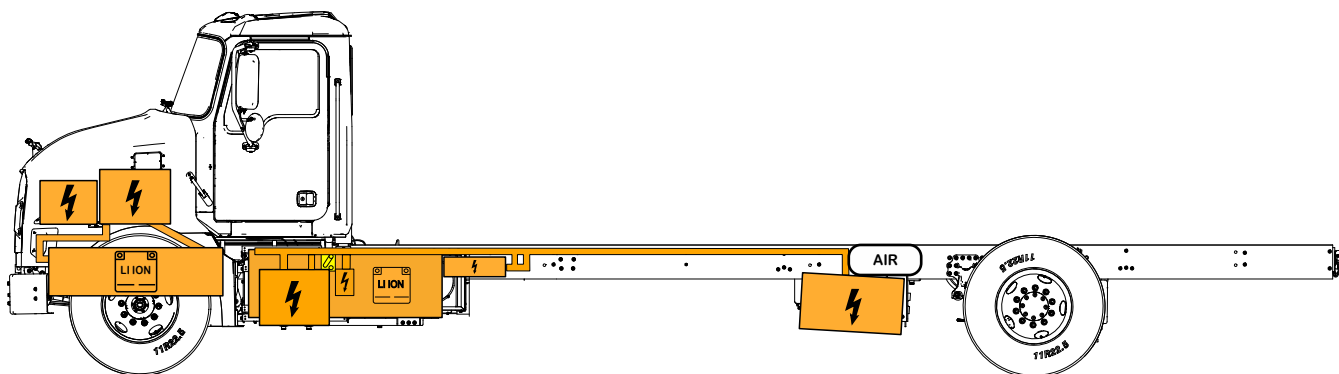




MACK TRUCKS

MDe6 / MDe7 (150 kWh battery pack)

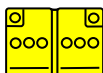
PRODUCTION START : 2024



High voltage
lithium-ion battery



Low voltage device
that disconnects the
high voltage



Low voltage battery



Air tank



Seat adjustment



Cable cut: Cutting
this cable disconnects
the high voltage.



Steering wheel tilt
control



High voltage
component



High voltage
cable



Gas strut,
pre-loaded spring



Ignition key /
Starter switch



Disconnect high
voltage device

NOTE

- (a) The above image shows information for one variant of this product. Number of axles, cab structure and number of traction batteries can vary depend on the variant of the product.
(b) These instructions do not cover the safety aspects of the components and equipments mounted by a third party (example: body builders).
(c) Discrepancy may exist in the translated documents as the original document is authored in English.

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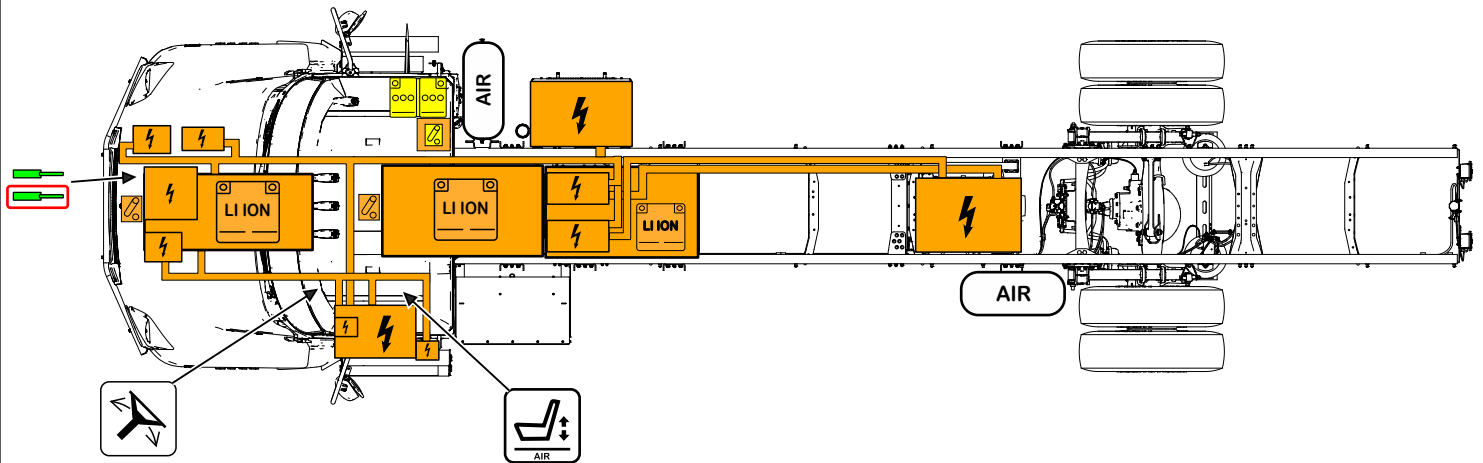
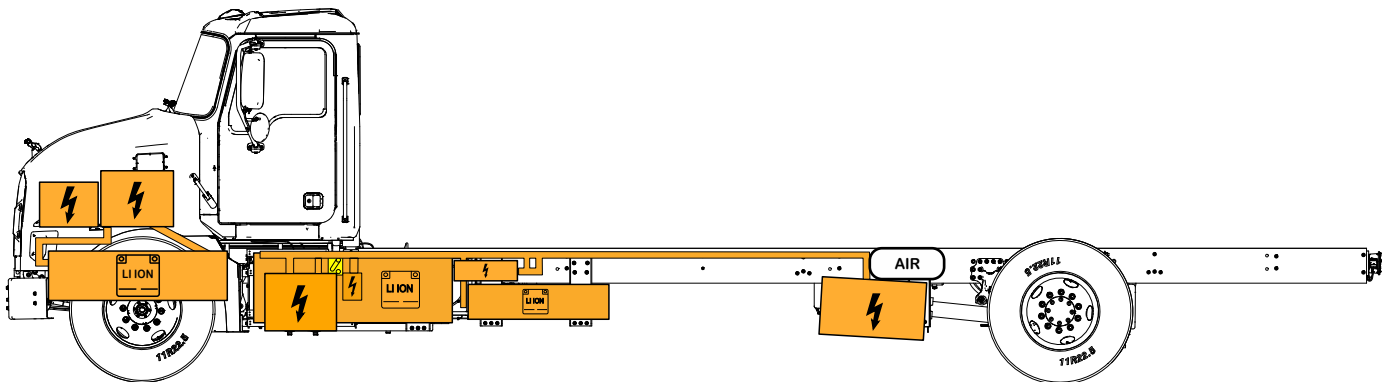
1



MACK TRUCKS

Mde6 / MDe7 (240 kWh battery pack)

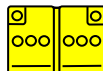
PRODUCTION START : 2024



High voltage lithium-ion battery



Low voltage device that disconnects the high voltage



Low voltage battery



Air tank



Seat adjustment



Cable cut: Cutting this cable disconnects the high voltage.



Steering wheel tilt control



High voltage component



High voltage cable



Gas strut, pre-loaded spring



Ignition key / Starter switch



Disconnect high voltage device

NOTE

- (a) The above image shows information for one variant of this product. Number of axles, cab structure and number of traction batteries can vary depend on the variant of the product.
- (b) These instructions do not cover the safety aspects of the components and equipments mounted by a third party (example: body builders).
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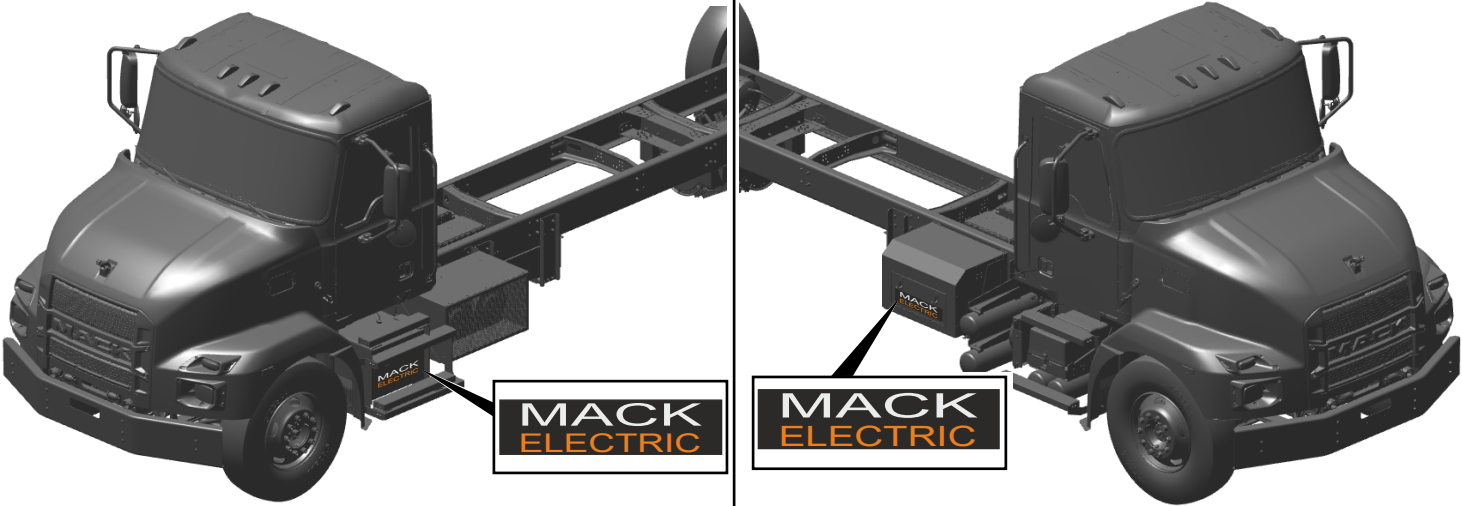
Version number

08/2024

Page number

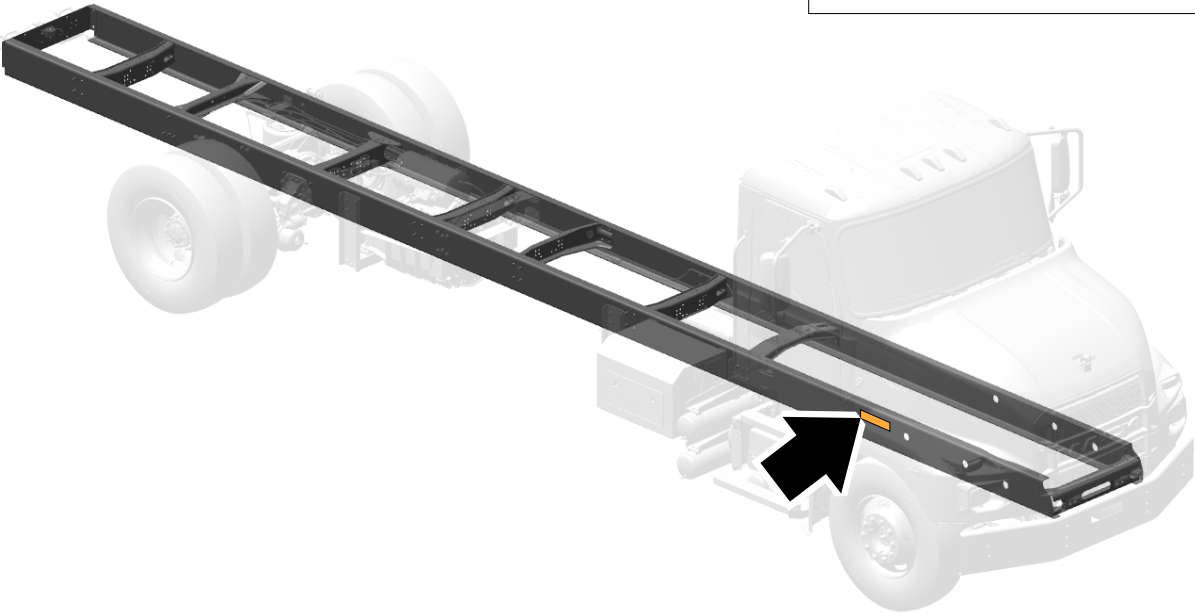
2

1. Identification/recognition



➡ Vehicle Identification Number (VIN)

1M2MDCAA4RSXXXXXX

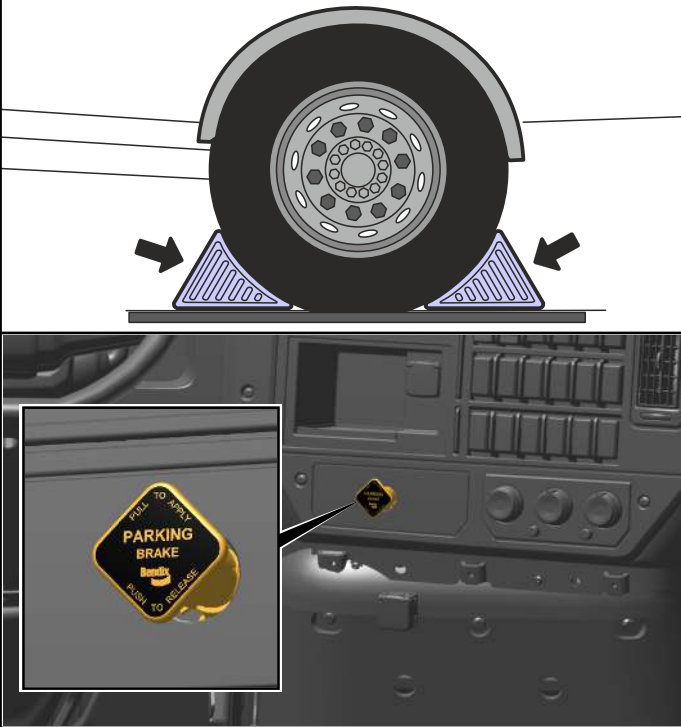


2. Immobilisation/stabilization/lifting



Always approach the vehicle from the sides to stay out of the potential travel path. Due to less noise it may be difficult to determine the state of vehicle.

- 1 Chock the wheels.
- 2 Apply the parking brake.

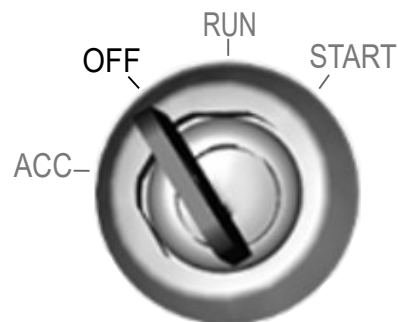


3. Disable direct hazards/safety regulations

➔ Primary procedure



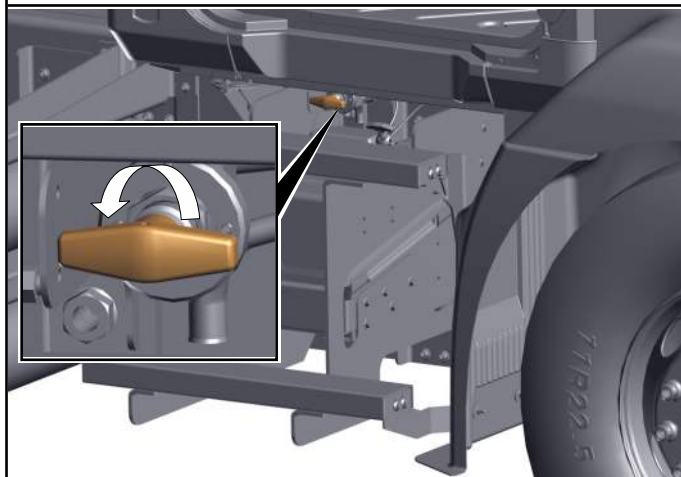
- 1 Turn off the starter switch and remove the key.



- 2 Turn off the isolation switch to initiate the high voltage disconnection process.

Note

The 12V isolation switch isolates power to the vehicle and the BMS. In an emergency do not rely on this switch to disconnect power to all systems.



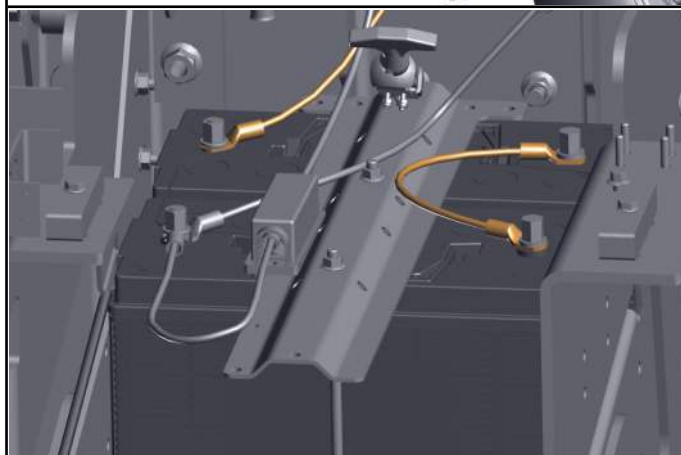
- 3 To fully disconnect power to the vehicle and the BMS, cut the cable between both the 12 voltage batteries on the right side of the vehicle as well as the cable from the negative post to the chassis. If time allows, these cables can be removed by loosening bolts with a wrench and removing by hand.

Note

Use insulated tools to cut the cable.

WARNING:

Always assume High Voltage (HV) is still live. Never touch, open, or cut any of the orange cables or components. Wait 10 minutes for the HV to deplete after 24V battery is disconnected.



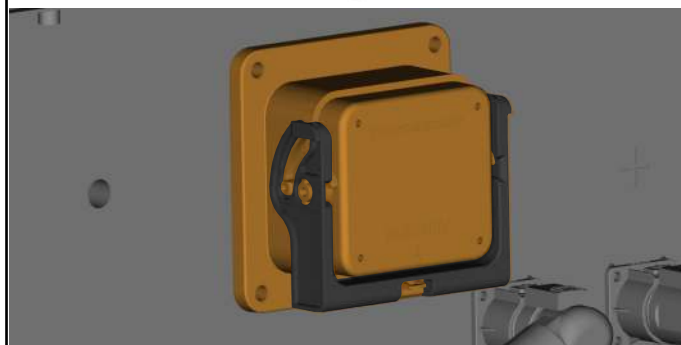
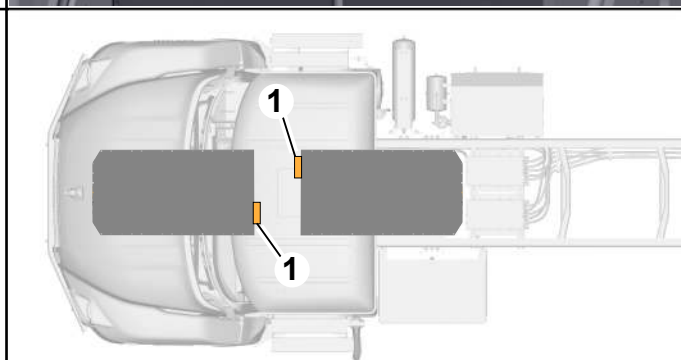
➔ Manual Service Disconnect (MSD) removal procedure for trucks with 2 battery pods (150kWh capacity)



- 1 Pull the red locking tabs and remove both MSDs (1) from the front 2 HV battery pods, as long as they are safely accessible.

Note

Removing Manual service disconnect (MSD) from any of the batteries will disable the traction voltage supply.



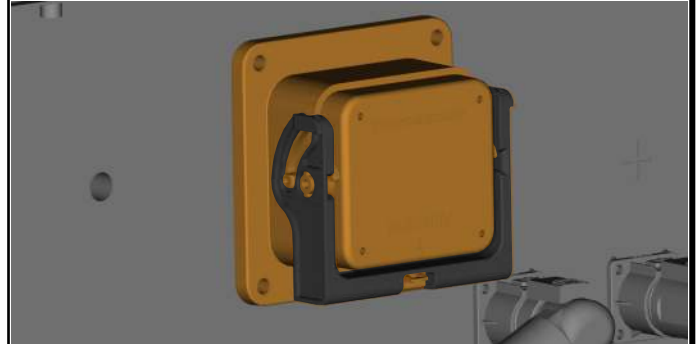
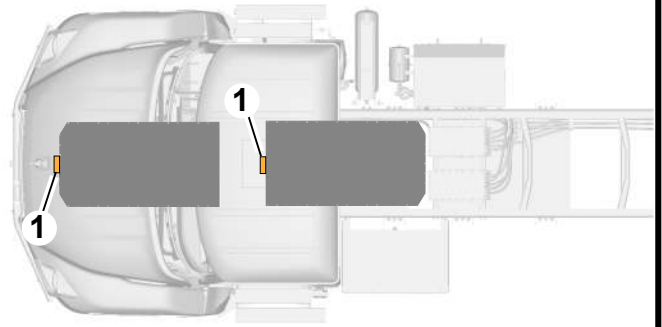
➔ Manual Service Disconnect (MSD) removal procedure for trucks with 3 battery pods



- 1 Pull the red locking tabs and remove both MSDs (1) from the front 2 HV battery pods, as long as they are safely accessible.

Note

Removing Manual service disconnect (MSD) from any of the batteries will disable the traction voltage supply.



➔ If unable to remove Manual Service Disconnect (MSD)

- 1 Remove or cut at least one of the following cables:
 - The small low voltage connector on the PDU labelled "Pack." The PDU is located under the hood, on top of the control tower structure.

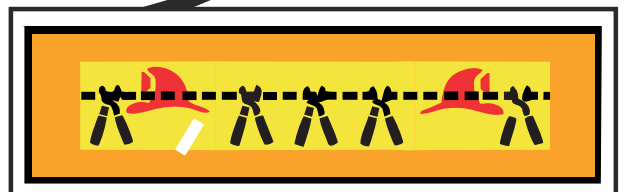
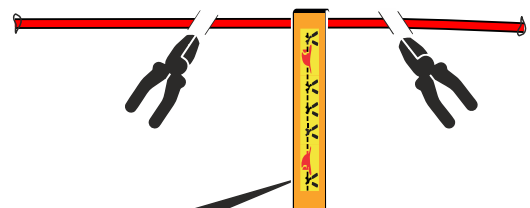


- The small low voltage connector on the front of the rear battery pod labelled "CAN."



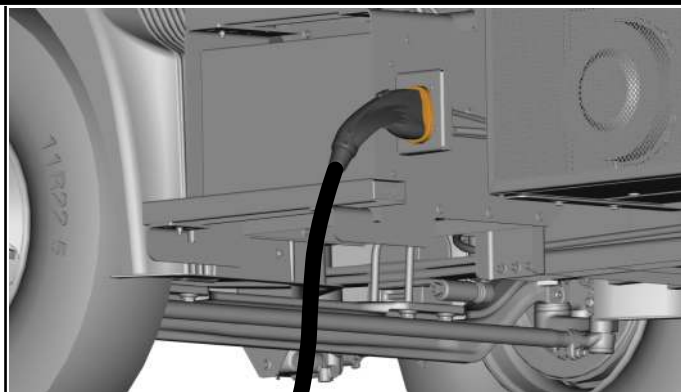
Note

Use insulated tools to cut the cable.



➡ If the truck is charging

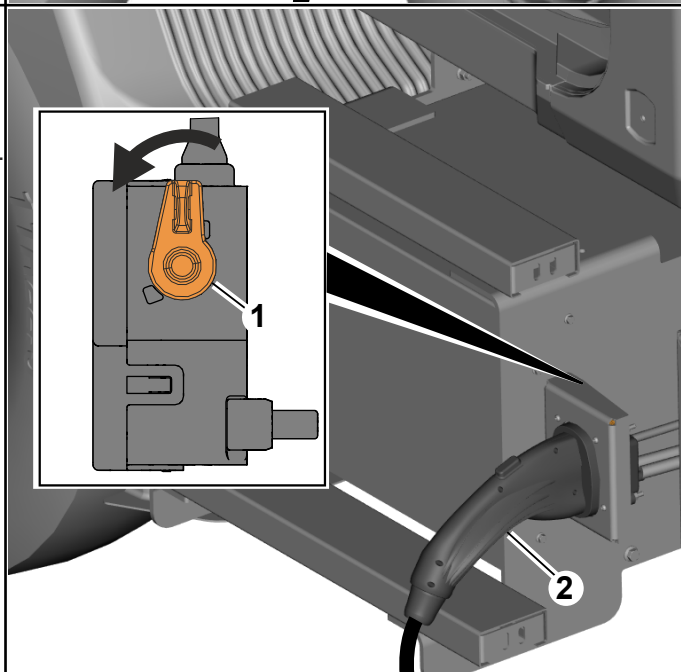
- 1 Disconnect the charging cable plug to unlock the charger.



➡ If the charging plug cannot be disconnected : retract the pin manually



- 1 Rotate the lever (1) and remove the charging plug (2).



4. Access to the occupants

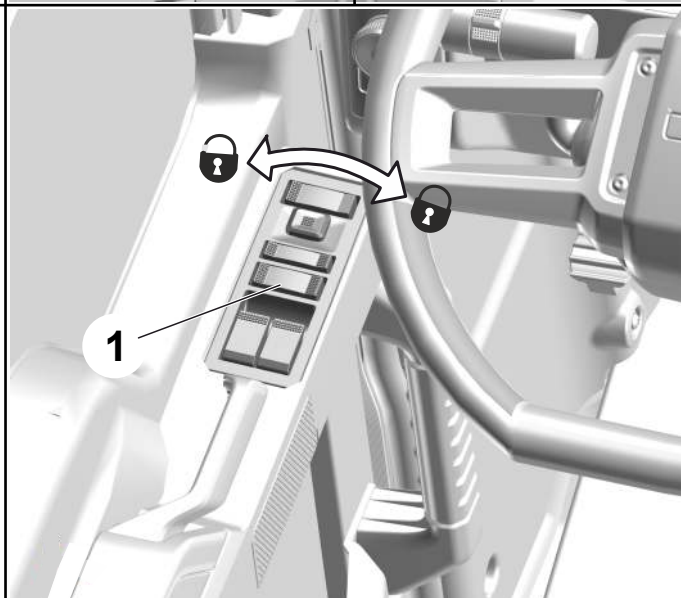
➡ Opening doors from the outside

- 1 To unlock the left-hand side door, turn the key to the unlock position (1) (Clockwise).
- 2 To unlock the right-hand side door, turn the key to the unlock position (2) (Counter Clockwise).

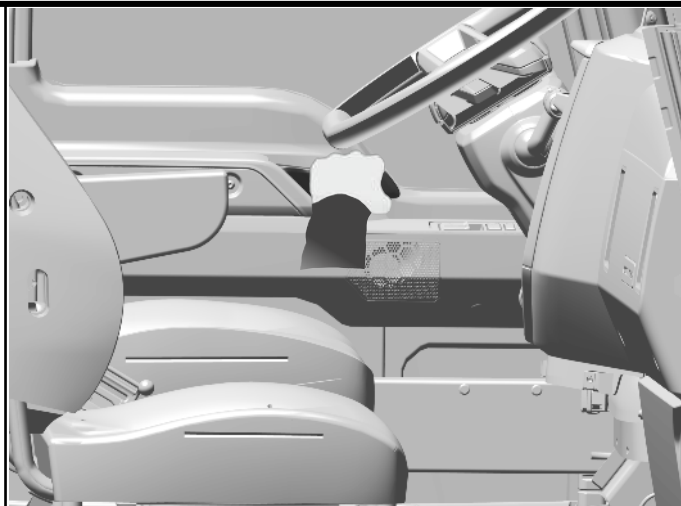


➡ Opening doors from the inside

- 1 To unlock doors from inside the cab electronically, push the power unlock switch (1) on the door control panel.



- 2** To open the door from inside the cab, grasp the handle and pull out while exerting some force on the door.



➔ Driver Seat adjustment



- 1** To adjust the seat height, push the switch (1).



- 2** Pull the lever (2) for sliding the seat to the desired position.



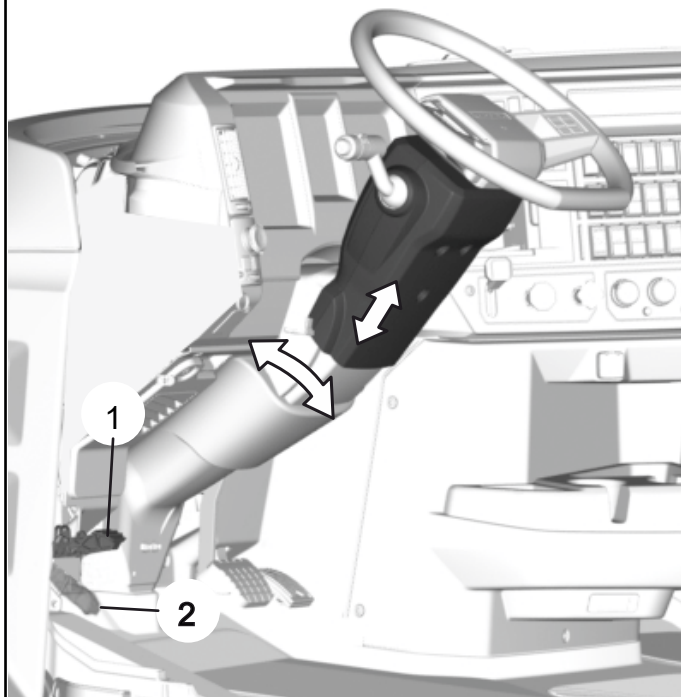
➔ Steering column adjustment



- 1** Press the pedal (1) all the way to adjust the telescopic length and angle of the column.

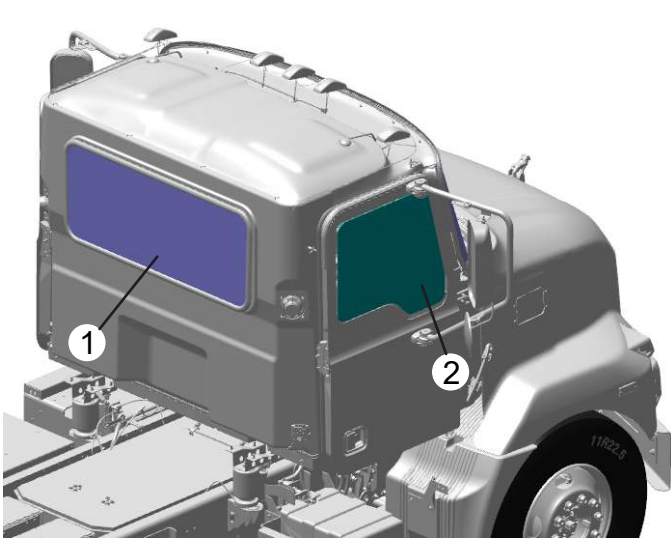
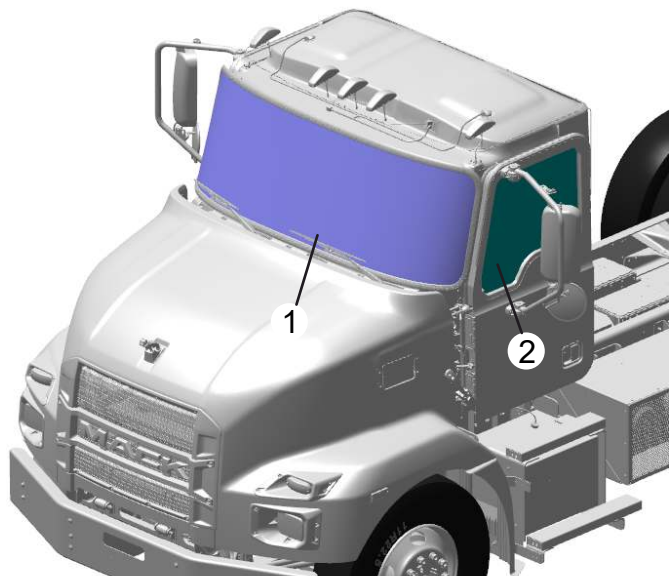


- 2** Press the pedal (1) to first position to adjust the angle of the steering wheel.



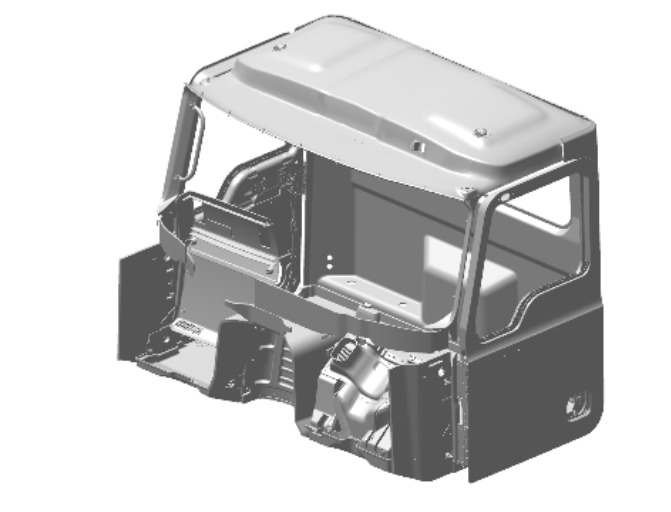
➔ **Windows and windshield**

Note
The windshield and rear glasses are made of laminated glass (1). Window glass are made of tempered glass (2).

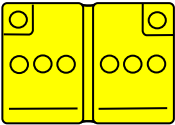
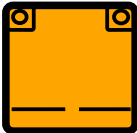


➔ **High strength zone**

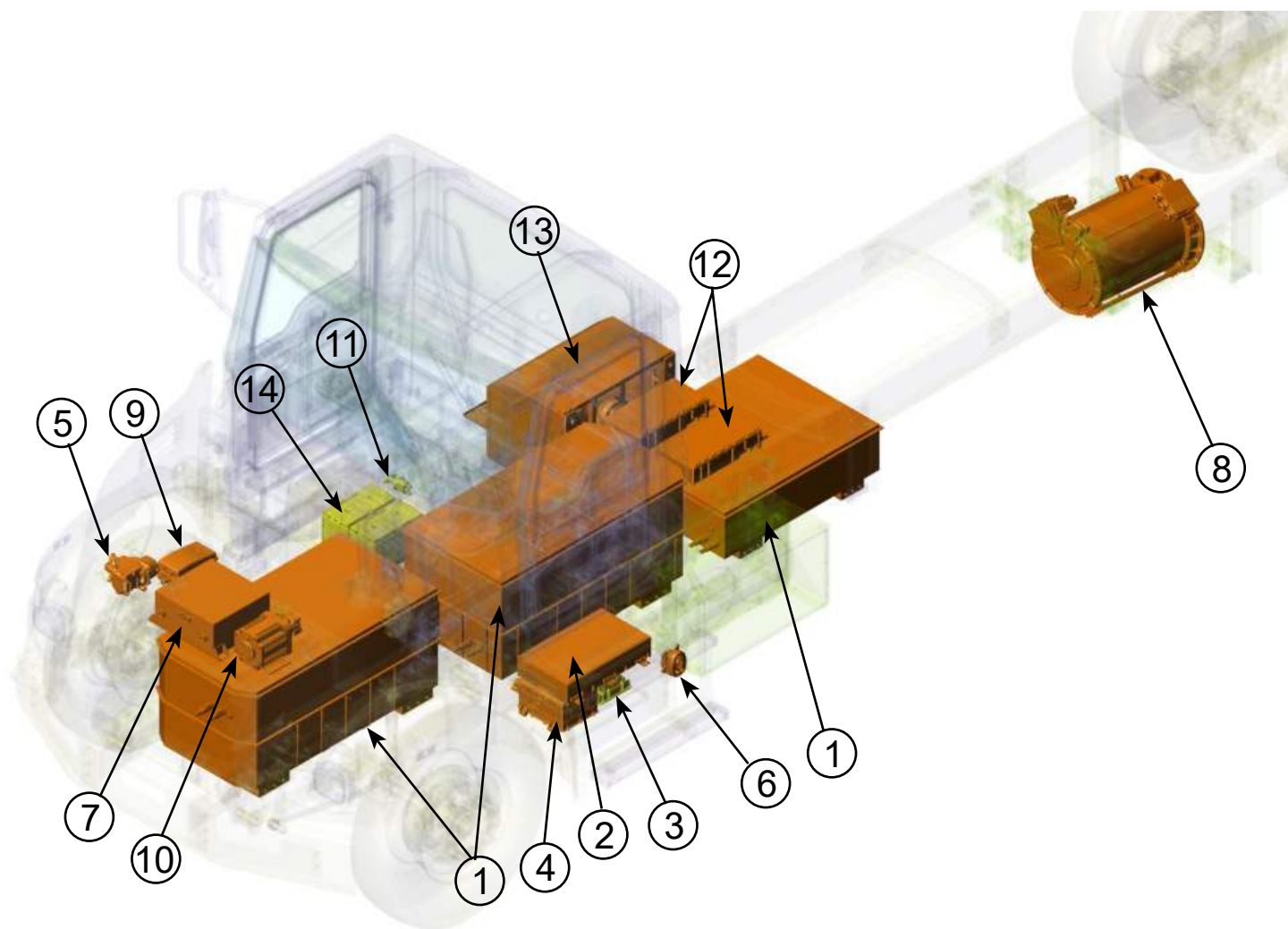
Note
There are no High-Strength and Ultra-High-Strength Steel in the cab. The cab structure is made predominantly of carbon sheet steel that does not exceed 340 Mpa.



5. Stored energy/liquid/gases/solid

	  	12 V
	     	600 V

➔ High-voltage component location



1. Traction Battery: 3 Li-ion battery pods supply maximum of 400 Volts and 300 Amps. Battery electrodes are made of Carbon, Lithium, Nickel, Manganese and Cobalt.

2. Charging Unit: 19.2kW AC charging. This device is used to convert AC power to DC power.

3. High Voltage (HV) V&T DC/DC Inverter: This device is used to charge 24V batteries during operation.

4. DC/DC Converter: This device is used to convert 24VDC to 12VDC.

5. AC Compressor: Provides compressed refrigerant to the vehicle interior environment.

6. Charging Inlet, CCS 1 Type: The CCS inlet is used for AC and DC charging. The CCS inlet is equipped with an actuator that prevents accidental disconnection of the charging plug on the truck side.

7. Power Distribution Unit (PDU): This device manages and monitors HV connections for the various HV components.

8. Traction Motor: Provides primary power to the drive train.

9. Heater: Provides heat to the vehicle interior environment.

10. Power Steering Pump: HV hydraulic motor and pump assembly provides hydraulic power to the power steering system.

11. 12V Isolation Switch & Battery Disconnect: This switch disables all power to the vehicle, including HV Traction Batteries.

12. Motor Control Unit (MCU): This device provides HV switching to drive the Traction Motor.

13. HV Air Compressor: This device produces compressed air to actuate the vehicles braking system.

14. LV Batteries: (2) 12V batteries are connected in series to create a 24V battery system.

6. In case of fire



Use a large sustained volume of water to extinguish lithium-ion battery related fire until, no heat can be seen coming from the battery pack with a thermal camera.



Note
Extinguishing a lithium-ion battery-related fire with water can produce hydrofluoric acid. Efforts should be made to control and collect run off water.



If other materials are involved, use class ABC fire extinguisher.



In case of thermal runaway, the lithium-ion batteries can release hydrogen fluoride and other toxic gases.



In case of thermal runaway, the traction battery can explode or breather valves (1) may emit large flames as a result of thermal runaway.



Personal Protective Equipment (PPE) of the first responders may be contaminated due to the exposed to hazardous chemicals.

7. In case of water submersion



The damage level of a submerged vehicle may not be visible.

Submersion in water can damage 12V, 24V and 600V components.

Handling a submerged truck without appropriate Personal Protective Equipment (PPE) may result in serious injury or death due to electric shock.

Avoid any contact with the traction voltage cables and electric components.

If possible disable direct hazards (Refer to “3. Disable direct hazards/safety regulations”).



8. Towing/transportation/storage



Check the condition of the lithium-ion batteries before towing. If the traction batteries are damaged, there is a risk of thermal or chemical reaction. It is recommended to take guidance from emergency response personnel before towing.

Delayed thermal event in the lithium-ion batteries can occur after they are damaged or after battery fire / heat suppression. A heat camera may be used to identify the thermal event.

To ensure the battery safety, it is recommended to:

- Park the electric truck involved in an accident in a suitable place maintaining a safe distance from other vehicles, buildings and combustible objects.
- Perform risk analysis based on the local situation. Observe the electric truck at least an hour.

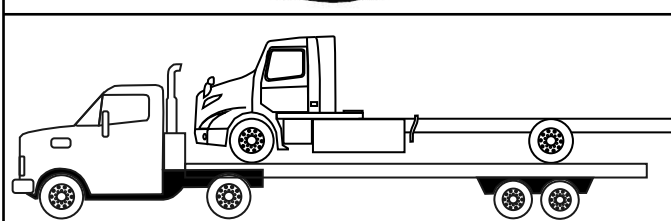
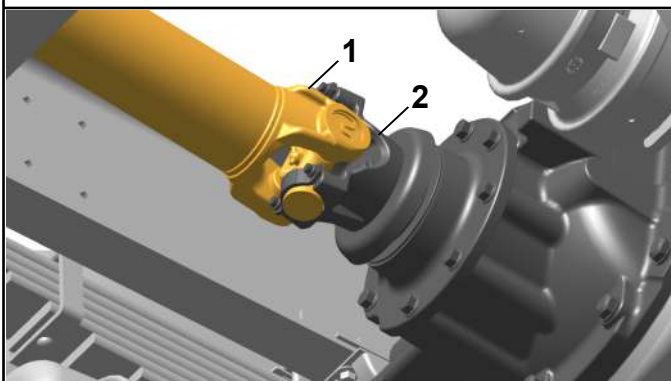


Before towing an electric truck with rear wheels on the ground, it is mandatory to disconnect the drive to the wheels.

The drive to the wheels is disabled by either remove the propeller shaft (1) from the driven axle (2) or by removing the drive shafts (3).

Note

Moving an electric truck with the rear wheels on the ground without disabling the drive may cause damage to the electric motor and the gearbox.



If it is not possible to tow the truck with rear wheels on the ground, it can be towed on a flat bed tow truck.

➔ Maximum loading during lifting and towing:

This information specifies the loading which can be applied when using towing hook, towing hitch cross-member, axles and torque stay anchorages.



Double towing hooks: Do not load each hook more than half the vehicle gross weight.

Maximum load on towing cross member are:

- Lengthways: 30 tons
- Vertically (lift): 7 tons
- Sideways: 17 tons

Note

When the vehicle is towed with the rear suspension lifted, the steering wheel must be locked with the steering lock.

Note

If roof deflectors cannot be removed, tow from the front of vehicle only.

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9. Important additional information



Do not cut any orange cables.

Do not touch any high voltage cables and electric components.

Do not perform any operation on a damaged truck without appropriate Personal Protective Equipment (PPE).

10. Explanation of pictograms used

	Use water to extinguish the fire
	Use ABC powder to extinguish the fire
	General warning sign
	Warning, Electricity
	Use thermal Infrared camera
	To indicate the risk of flammability
	To indicate the risk of an explosion
	To indicate the risk of corrosive material/substances
	Hazardous to the human health
	To indicate the risk of acute toxicity